If the primal is a minimization problem, and the dual is written as a maximization problem, then the following rules apply.

- For each constraint of the primal there is a dual variable. The sign of the dual variable depends on the sign of the primal constraint, according to the following table:

Constraint of the primal	Dual variable
$=$	free
$\leq$	≤ 0
$\geq$	≥ 0

- For each primal variable there is a dual constraint. The sign of the constraint depends on the sign of the primal variable, according to the table below.

Primal variable	Dual constraint
≥ 0	$\leq$
≤ 0	$\geq$
free	$=$

Write the dual of the following linear optimization problem in two ways: first use the summary tables above, and then use the Lagrangian method in order to see the correctness of those summary tables.

$$\min x_1 - 4x_2 + 2x_3$$

subject to

$$\begin{aligned} 2x_1 + x_2 &\geq 10, \\ x_1 + 4x_3 &= 16, \\ x_2 - x_3 &\leq 5, \\ x_1 &\geq 0, \\ x_2 &\in \mathbb{R}, \\ x_3 &\leq 0. \end{aligned}$$